Deep Learning Approaches for Brain Tumor Detection Using Hyperspectral Imaging

[1]Garima Jaiswal*, [2]Anika Sharma, [3]Mannat Aggarwal, [4]Eshit Saini

[1,2,3,4]Bennett University, Greater Noida, India

[1]garima.jaiswal@bennett.edu.in*
[2]anikasharma0398@gmail.com
[3]mannataggarwal07@gmail.com
[4]saini.eshit@gmail.com

***Abstract*— Detection of brain tumors in their early stages is crucial for improving patient outcomes and treatment efficacy. Many related pragmatic clinical tools and various Machine Learning models are made for the effective diagnosis of patients, but they still provide low accuracy scores. Therefore, the need to develop a more precise model for better screening of patients to detect tumors still exists. There has been a growing interest in using methodologies like Convolutional Neural Networks (CNNs) in medical imaging in recent years. By applying these advanced techniques alongside various machine learning (ML) algorithms, this study aims to discover the model which yields the best accuracy for detecting brain tumors.**

Keywords*— *Brain tumor detection, CNNs, hyperspectral imaging, tumor screening, classification

I. INTRODUCTION

Cancer is caused by the transformation of normal cells into tumor cells. They grow uncontrollably, forming masses called tumors, which destroy healthy tissue. In terms of mortality, the Central Nervous System and Brain Cancers ranked tenth in the year 2020, with an anticipated 308,102 incident cases that were found to be linked to about 251,329 deaths globally. In 2040, these figures are expected to rise by 38.5% for occurrences and a whopping 43.7 % for death [1]. The most common type of brain tumor in adults is glioma. As the most aggressive glioma kind, it does not allow for long-term survival among patients [2]. Therefore, early diagnosis is essential for improving patient outcomes and directing successful treatment plans. Although Magnetic Resonance Imaging (MRI) has been widely used to detect tumors, its diagnostic accuracy is limited because it frequently requires more spectral resolution to identify minute variations in tissue composition [3]. In India, Central Nervous System (CNS) and brain cancers present a significant medical health obstacle, with statistics showing an alarming rise in cases and mortality rates over the years. About 32,574 new cases of brain tumors and CNS cancers were reported in 2020, making up roughly 2.3% of all cancer cases in the nation [4]. This high occurrence rate goes to show the growing number of brain tumors in India's healthcare system [5]. Current intra-operative techniques for brain tumor surgery primarily aim towards improving the precision of tumor removal while minimizing the potential harm to the surrounding healthy tissue.

Incorporation of neuronavigation tools such as Intraoperative Magnetic Resonance Imaging (iMRI) [6,7], Intraoperative Ultrasounds (IOUS) [8], neuronavigation tools, and the use of markers like 5-aminolevulinic acid (5-ALA) [9], significantly impact the intervention technique, allowing neurosurgeons to work with safer and more efficient methods. While each of these approaches has enhanced the safety and effectiveness of brain tumor resections, they all come with drawbacks. To overcome the shortcomings of the current methods, there is an increasing interest in exploring non-invasive technologies such as Hyperspectral Imaging (HSI), which provides real-time tumor diagnosis without the need for invasive procedures or specialized surgical tools. Hyperspectral Imaging (HSI) has the potential to improve intraoperative operations significantly. This would lessen the need for invasive demonstrations frequently sent to pathology for analysis during surgery. The traditional approach prolongs surgery and delays decision-making because the results are only sometimes definitive and immediate. Real-time HSI may offer a more accurate differentiation between tumor and normal brain tissue [10], particularly for low-grade tumors that are challenging to distinguish, as surgeons today primarily rely on visual cues to differentiate between healthy brain tissue and malignancies. As Hyperspectral Imaging (HSI) allows for recording biochemical and morphological changes in tissues, it has been extensively utilized in medical diagnostics [11] and oncology to diagnose several cancers, such as skin, colorectal, and breast cancers [12]. Hyperspectral Imaging (HSI) consists not only of the visible spectrum but also the infrared and is, thus, an advanced imaging method that captures a wide range of light with various wavelengths. Compared to traditional RGB photographs, which use just three-color channels (red, green, and blue) to display scenes, HSI records hundreds of wavelengths in many spectral bands at each pixel.

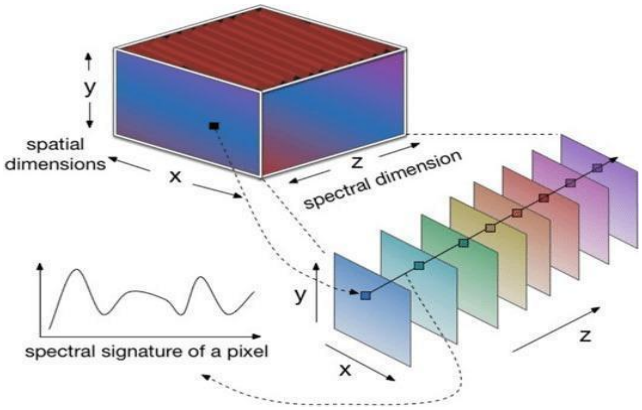


Figure 1: HSI cube

As shown in Figure 1, a three-dimensional hyperspectral data cube can be created that includes both spatial and spectral information about the object being scanned. The data cube consists of a spectral dimension that stores the reflectance of light at different wavelengths at each location on the item and two spatial dimensions that represent the object's surface.

Convolutional Neural Networks (CNNs) can recognize significant features in visual data and are broadly used for image classifications. This enables CNNs to process the hyperspectral data cube in a way that combines spectral information, which is crucial for detecting biochemical differences with spatial structure and helps differentiate tissues in imaging. Hence, they can distinguish essential features in the composition of tumors by automatically extracting both spatial and spectral characteristics from hyperspectral images [13]. Each pixel in the hundreds of spectral bands that make up HSI data contains spectral information that describes the characteristics of tissue reflectance. According to research, CNN-based models perform more accurately than conventional methods when classifying brain tumors from HSI images [14,15].

Support Vector Machines (SVM) are often employed to classify HSI data, especially for spectral classification tasks related to detecting brain tumors. Finding the best hyperplane to divide various groups (such as tumor vs. non-tumor tissue) based on their spectral signatures is the aim of high-dimensional datasets like hyperspectral images, which is where SVMs excel. In detecting brain tumors, Support Vector Machines have proven to facilitate tissue type classification by analyzing the spectral properties of the tissues. Even without considering spatial information, SVMS can achieve great accuracy with carefully selected spectral features. Studies have shown that SVM models effectively distinguish tumor margins, highlighting their utility in delineating them [16]. This paper aims to present a novel approach to real-time brain tissue classification using hyperspectral imaging (HSI) combined with Convolutional Neural Networks (CNN). Our primary objective with this research was to enhance the accuracy and efficiency of in-vivo brain tumor detection for further surgical use. The main contribution was the employment of CNN models which are well-suited for extracting spatial features from hyperspectral images.

II. LITERATURE REVIEW

There have been numerous attempts to integrate HSI into machine learning algorithms for better detection. While this is in its initial stage, it has shown promise for the future.

Tejasree et al. [17] and Urbanos et al. [18] implemented a method that infused HSI data with ML models like support vector Machines, Random Forests, and Convolutional Neural Networks to classify tissue types in brain cancer images. By applying detailed preprocessing, such as spectral correction and normalization, the classification was expected to be improved further. However, it encountered difficulties distinguishing tissues with similar spectral signatures, leading to ever-changing results and overfitting training data, limiting the model's reliability for practical application. This meant there was a need for more expansive models, but that would mean compromising on unique identification. This was made better by Leon et al. in [19], [20] by using VNIR (Visible and Near-Infrared) and NIR (Near-Infrared) sensors to capture in-vivo brain images, aiming to identify spectral differences between normal and tumor tissues. By combining images from both VNIR and NIR ranges into a single-dimensional cube, this method expanded spectral coverage, thereby increasing accuracy. However, using two cameras and the genuine requirement for accurate images increased complexity, increasing cost and reducing real-time application. This addressed the issue of spectral overlap, but the setup and cost negatively compensated for the advancements.

Another approach by Giannantonioa et al. [21] was a multi-step framework, a multi-scale CNN for feature extraction, graph-based clustering for feature selection, and gradient boosting for ensemble classification. This technique effectively captured the rich data within HSI, grouping similar features to reduce redundancy. However, this added significant computational overhead, making it difficult to employ in real-time. It also lacked generalization based on in-vivo data used. Additionally, its similarity to working as a 'black box' reduced its chances of acceptance since the rationale behind it is very important to practitioners.

These studies by Leon et al. [22] highlight several recurring challenges across HSI and ML-based brain tumor classification research. High computational demands, particularly for models using advanced ML techniques, are a daunting barrier. Overfitting problems occur regularly, and making the model generalized is a task. The nature of HSI's availability proves to be an issue as well. An approach maintaining generalization cuts back on precision and vice-versa. Sancho et al. [23] also

conducted a related study, which developed an Augmented Reality (AR) system, which integrated real-time acquisition and processing for classification mapping. The HSI in-vivo brain tissue dataset was used here with the utilization of advance preprocessing which included steps like Hyperspectral (HS) cube transformation and spectral correction. The system was able to achieve a processing speed of 14 Frames Per Second (FPS), which made it suitable for real-time applications. Despite promising outcomes, the study was unable to conclude fully due to some level of confusion between healthy and tumor tissues. On the other hand, Kifle et al. [24] solely focused on pediatric brain tissue segmentation with the use of a Snapshot Hyperspectral Imaging (sHSI) camera. These were processed with the assistance of Machine Learning (ML) classifiers. They made use of MATLAB's ImageSegmenter for ground-truth-based image splitting and evaluated the performance of Random Forest (RF) classifiers with 10 estimators against SVM. However the study's reliability on a small dataset and low sensitivity for tumor segmentation limited its wider applicability.

The high cost and complexity of systems that require multiple cameras or intricate image registration further limit their real-time feasibility, as these setups are often bulky and challenging to manage. Finally, while these models achieve high classification accuracy, their complexity can make it difficult to interpret their decisions—an issue that affects trust and the eventual modeled adoption. HSI and ML integration holds significant potential for improving brain tumor detection, but they come with problems that must be solved head-on. Table 1 gives an overview of the methods used so far and their effectiveness in detecting brain tumors.

Paper ID	Pre-processing	Dataset	Model	Accuracy	Limitations
Tejasree et al [17]	Image Calibration, Noise Filtering, Spectral Correction, Data Normalization	HSI In-vivo brain tissue	Gradient Boosting; Multi-Scale CNN	SVM: 99.8% to 100% RF: 100% MCGC: 99.9% to 100%	Data complexity; Disruption due to Noise; limited execution.
Urbanos et al [18]	Hsi Cube Creation, Calibration, Spectral Correction, Data Normalization	13 In-Vivo HSI from Patients	SVM, Random Forest, CNN	RF:95%; CNN: 91.9%; SVM: 74.6%	Limited Dataset Size Limited Complexity Accuracy Variability
Leon et al [19]	Calibration Soothing Band Removal Registration	HSI In-vivo brain tissue	SURF and MSER detectors	SVM-VNIR: ~85%, NIR: ~82%, Fused: ~92% RF-VNIR: ~88%, NIR: ~85%, Fused: ~94% KNN:VNIR: ~83%,NIR: ~81%, Fused: ~90%	Limited Dataset Size Limited Spectral Analysis
Leon et al [20]	Image calibration, Band removal, Upsampling (bilinear interpolation), Region of interest (ROI) cropping	HSI Registration Dataset, HSI Spectral Reference Dataset, HSI Plastic Dataset, HSI Brain Dataset	fused VNIR–NIR HS data, K-medoids, Hierarchical K-means, SVM, RF, KNN	Fused VNIR–NIR data increased classification accuracy by 21%	a small dataset lacks labeled tumor pixels. validation is needed with larger datasets, with more diverse tumor types
Giannantonioa et al. [21]	Normalization, noise reduction, dimensionality reduction enhance reflectance values. Improving signal-to-noise ratio through the SLIC algorithm.	18 In-Vivo HSI from Patients	RF, SVM, MLP	RF = 86% SVM(RBF)= 91% MLP classifier= 92%. 104-to-3 channels: 76-81% 104-to-6 channels: 82-87% 104-to-12 channels: 86-89% NN with 12 most important channels= 81% accuracy (94% precision, 80% recall).	small dataset, Image quality sensitive to Inter-patient variability
Leon et al [22]	Noise reduction, Normalization to reduce data dimensionality	In-Vivo HSI brain cancer dataset	random forest (RF), KNN (Euclidean and Cosine), SVM (SVM-L and SVM-RBF), DNN.	F1-score= 70.2% ±7.9%. SVM-RBF accuracy=91.5% ± 4.7%, DNN tumor tissue sensitivity= 65.9%	Variable patient data, difficulty capturing high-quality images; accuracy degraded if acquisition conditions are not optimal
Sancho et al. [23]	HS cubes transformation, image calibration, spectral correction	HSI In-vivo brain tissue	SVM with RBF kernel, K-Means clustering.	Area Under the Curve (AUC): 95%,	(14 FPS exposure time), confusion between healthy and tumor tissues can reduce subjective quality.
Kifle et al. [24]	Ground truth based splitting; MATLAB ImageSegmenter	60 RGB images, 234 visible HSI images,60 infrared HS images	RF with 10 estimators; Performance compared to SVM.	RGB: IoU of 0.76, visible, infrared HSI IoUs of 0.57 and 0.59,	Small dataset, low sensitivity, particularly for tumor segmentation

Table 1: Literature review

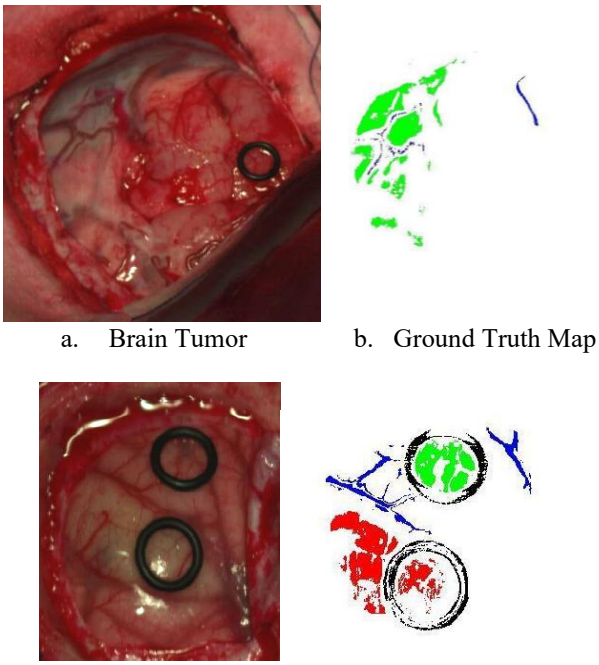
III. METHODOLOGY

This section describes the detailed process of utilizing hyperspectral imaging for brain tumor detection. The methodology comprises the details for the dataset preparation, preprocessing, augmentation, and ML algorithms.

2.1 Dataset Description

For this study, we have used an established, public dataset of Hyperspectral images from the NPJ Oncology Database in the NPJ Oncology Journal [25] which consists of 70 HSI data points that amount to 11.2 GB of unextracted data. This dataset has been extensively used in prior research.

We excluded several images from the database as they didn't contain the relevant tumor pixels due to unreliable labeling or the patient undergoing surgery for another pathology. After excluding these images, the study procedure finally included 61 HS images from 34 different patients. All these images were acquired intraoperatively [26,27].
It includes calibrated HSI images and their corresponding labels in NumPy arrays, their associated dark and white reference images, and the ground truth maps for every patient. Figure 2 shows the RGB images of tumors with their corresponding ground truth maps. The (ground truth) GT maps are composed of four classes: TT (red color), NT (green color), BV (blue color), and BG (black color). White pixels in the GT maps represent the non-labeled pixels because only the pixels that had high confidence of belonging to a particular class were labeled [22,28]



a. Brain Tumor b. Ground Truth Map

c. Brain Tumor d. Ground Truth Map

Figure 2: Brain Tumor images and their corresponding Ground Truth maps

Table 2 gives an overview of the clinical statistics and the patient demographics. With a median age of 51.5 years, the study consists of 21 males and 13 females aged 30 to 73. Out of these 34, about 82.4% of the people (28) had a primary tumor. The most common primary grade tumor was G4(44.1%) in 15 people, followed by G1 and G2(14.7%) in about five people each. The other three were classified to have G3 tumors (8.8%). The rest of the tumors (17.6%) were secondary, from breast carcinoma in 3 people to renal carcinoma in the kidneys and 2 in the lungs, an adenocarcinoma and carcinoma. The majority of the tumors were found in the right temporal lobe, about 23.5%, followed by 20.6% in the frontal and right parietal lobes in 7 people each.

Variable [patients with no missing values/Total patients]	Characteristic	Total (n)	%
Sex [34/34]	Male	21	61.8
	Female	13	38.2
Age [33/34]	Median	51.5	-
	Range	30–73	-
Tumour Type [34/34]	Primary	28	82.4
	Secondary	6	17.6
Primary Tumour Grade [28/34]	WHO Grade 1	5	14.7
	WHO Grade 2	5	14.7
	WHO Grade 3	3	8.8
	WHO Grade 4	15	44.1
Metastasis [6/34]	Breast	3	8.8
	Lung	2	5.9
	Kidney	1	2.9
Location [34/34]	Right Frontal Lobe	3	8.8
	Left Frontal Lobe	7	20.6
	Right Parietal Lobe	7	20.6
	Left Parietal Lobe	4	11.8
	Right Temporal Lobe	8	23.5
	Left Temporal Lobe	1	2.9
	Right Occipital Lobe	2	5.9
	Left Occipital Lobe	1	2.9
	Cerebellum	1	2.9

Table 2: Summary of patient demographic and tumor characteristics

2.2 Data preprocessing

The NPJ database was prepared meticulously for hyperspectral imaging analysis. Numerous necessary preprocessing steps and strategies have been implemented to improve the data quality, as shown in Figure 3, facilitating practical model training. Initially, the raw hyperspectral images were radiometrically calibrated, where the dark and white reference HDR files were used to correct sensor responses and illumination variations. This calibration step ensures that the spectral data represents the entirety of the underlying tissue characteristics as accurately as possible, keeping samples consistent, particularly when examining minor spectral variations critical to tumor identification.

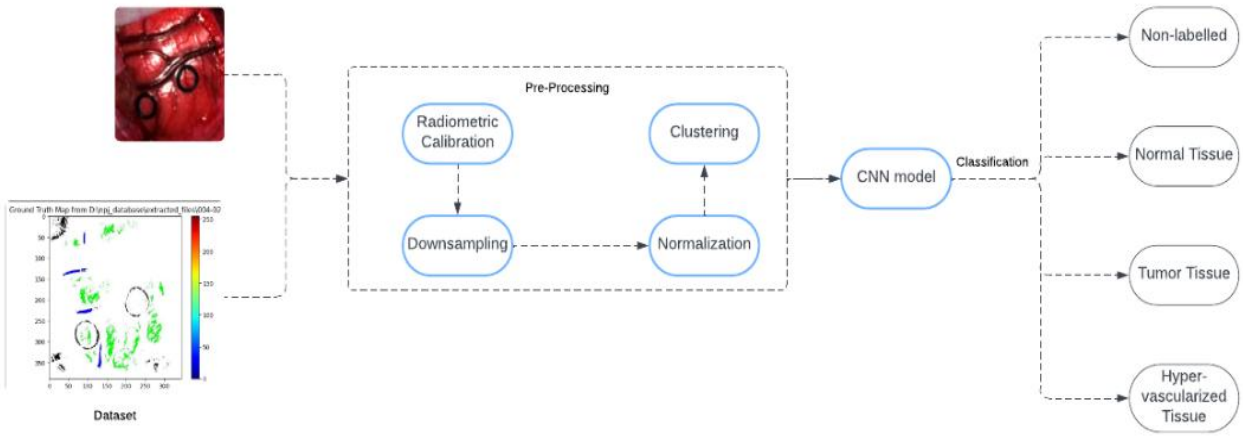


Figure 3: Flowchart of Methodology

Now, to standardize the input dimensions for the CNN, the images were resized to keep a consistent dimension of 512x512 pixels. This resizing is vital for batch processing and overall model functioning. Due to the high dimensionality of HSI data, a reduction was made with Principal Component Analysis (PCA) to combat that. This assisted in reducing noise and keeping essential features relevant to tumor identification. Padding was used to recognize any further discrepancies to ensure that all ground truth maps were appropriately aligned with the resized images, thus maintaining the integrity of the spatial information during analysis. Given the limited size of the dataset, data augmentation was also incorporated as a preprocessing step to improve the model's generalizability and robustness. Techniques like rotation, flipping, and scaling were also used to increase variability and navigate potential overfitting, which arose from training on small datasets [29][30].

2.3 CNN Architecture

Before the use of CNN, various methods were tested to reach the best accuracy possible. traditional machine learning methods involving random forests, decision trees and support vector machine achieved relatively lower accuracy. Application of relatively complex method of EfficientNet led to overfitting of the data. However, this was not the case with CNN, hence, the decision was made to use and finetune the CNN model which led to the following architecture as shown in Figure 4.

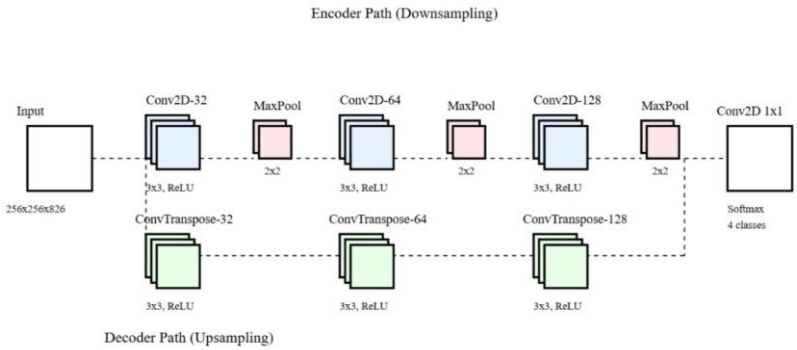


Figure 4: CNN Architecture

Figure 4 gives a visual representation of the CNN architecture that has been implemented. Three Convolutional Layers with kernels of size 3x3 were used, and the number of filters was progressively increased from 32 to 128. Each layer uses the ReLU activation function and applies padding to preserve the spatial dimensions of the input data. MaxPooling Layers are used after each Convolutional layer with a pool size of 2x2 so that the spatial dimensions of the feature maps are reduced in half, but the essential features are not lost. Upsampling layers were applied to recover the original spatial dimensions. We used three transposed convolutional (upsampling) layers, starting from 128 filters and decreasing to 32 filters. These layers also use the ReLU activation function and are configured with strides of 2, doubling the spatial dimensions after each layer. The output layer is used at the end with kernels of size 1x1, which are used to produce output for each of the five classes using the SoftMax activation function as illustrated in Figure 5.
Class probabilities for each image pixel were employed for multi-class classification in this layer. These classes are Not Labeled (Class 0), i.e. pixels that are ambiguous in their class type; Normal Tissue (Class 1), which are healthy tissues; Tumor Tissue (Class 2), where pixels represent the high-probability areas of having been affected by cancer, Hyper-vascularized Tissue (Class 3) with the pixels representing relatively probabilistic areas of having been affected by cancer or other conditions and finally, Background (Class 4), which has irrelevant pixels representing the image's background -not belonging to any tissue type.

2.4 Results

The model's performance was concluded with a training accuracy of about 77.17% and a test accuracy of about 74.85%, slightly lower than the training accuracy. Despite the gap between the accuracy of the train and test data, these results show that the model has basic differentiation capabilities and can effectively generalize to the test data by identifying malignant tissues in the hyperspectral images.

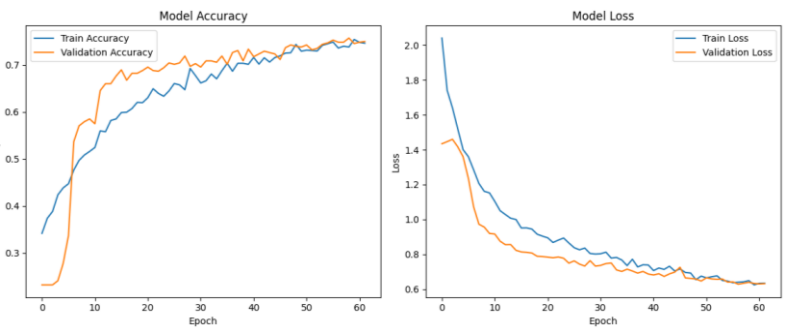


Figure 5: Model Accuracy Plot

In the plots in Figure 5 above, we conclude that for the Model Accuracy graph on the left, the training accuracy starts higher and gradually increases, indicating that the model is steadily learning on training data. Validation accuracy starts lower than

initial training accuracy but increases immediately before steadying out; it stabilizes before training accuracy. This shows that the model generalizes unseen data to a certain extent. For the Model Loss graph on the right, training and validation show similar losses. That means they both show a steady decrease with an increase in the number of epochs. The close alignment of the curves also indicates no overfitting within the model.

Paper ID	Dataset	Approach	Accuracy (%)
Present Study	NPJ Oncology Database	HSI and CNN model	77.17
[31]	PIONEER diagnostic centre(brain tumor and normal brain)	MRI along with ML algorithms	70
[32]	Harvard Medical School (http:// med.harvard.edu/AANLIB/)	NON HSI and multiple ML algorithms	74.16 to 97

Table 3: Comparative Analysis

A comparative analysis of the results is shown in Table 3; we found that where our methods achieved an accuracy of 77.17%, Mahajani et al. [31] reported an accuracy of 70%. They adopted traditional machine learning techniques on MRI images, which resulted in lower accuracy and highlighted the effectiveness of deep learning techniques over older ones. By employing multiple machine-learning approaches, Lavneet et al. [32] achieved an accuracy ranging from 74.16% to 97%.
While Mahajani et al. [31] relied on only machine learning algorithms, Lavneet et al. [32] highlighted a variability of accuracy, which shows a reliance on specific approaches over others. Our model consistently shows promise for broader applicability and can be further enhanced with multi-modal imaging or advanced architectures like transformers.

IV. Conclusion

In summary, this study shows how integrating convolutional neural networks with HSI results in the early identification of brain cancers, especially aggressive ones like grade IV. This combination technique assures a non-invasive method for accurate diagnosis of brain cancers. To normalize the HSI data and make the model more precise, we have applied various preprocessing techniques, radiometric calibration, and Principal Component Analysis for dimensionality reduction.
The results of our study demonstrated the potential of machine learning techniques in the ever-evolving field of medical imaging, especially in diagnosing brain tumors, where quick action can be life-saving. When traditional techniques and MRIs reach their limits, integrating a hyperspectral imaging approach with CNN-based analysis dictates a strategy that could help enhance surgical precision and diagnostic accuracy, which, in turn, could lead to improved patient outcomes.

V. Future Scope

The combination of HS Imaging and Convolutional Neural Networks opens the door to a solution with a lot of promise for detecting brain cancer, particularly glioblastomas. Subsequent work on better models for the fusion of spatial and spectral data could be immensely beneficial for future research. It could be further built on by applying complex structures like Transformers and 3-dimensional Convolutional Neural Networks, which could also aid in the fusion. Real-time applications are a much-needed topic to be researched.
A further step to achieve better diagnostic precision could involve multi-modal imaging techniques. The combination of HSI with magnetic resonance imaging (MRI) would integrate the spectral richness of HSI with the detailed structural information provided by MRI. Additionally, incorporating functional imaging data, such as positron emission tomography (PET) or functional MRI (fMRI), could create a comprehensive diagnostic tool that further captures biochemical, morphological, and functional changes in tissues. Such a system would set new standards in diagnostics, offering unprecedented accuracy and detail in identifying and characterizing tumors. This progress is not limited to brain cancer detection alone. The progress can be extrapolated to other detection mechanisms with precise margins, such as head and neck, liver, lung, and all other kinds of cancers and cases that require specific tissue classification.

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